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Research article

Acceleration tracking control combining adaptive control and off-line compensators for six-degree-of-freedom electro-hydraulic shaking tables

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ABSTRACT

An electro-hydraulic shaking table (EHST) is an essential experimental facility to simulate in real-time actual vibration situations. An adaptive controller combined with off-line compensators is proposed to improve the acceleration frequency bandwidth and tracking accuracy of a six-degree-of-freedom (6-DOF) EHST. A servo controller has been employed to implement acceleration closed-loop and coordinate control of the 6-DOF EHST. A recursive extended least-squares algorithm is employed to identify acceleration closed-loop transfer functions and a zero-phase-error tracking controller is used to design off-line inverse model compensators using the identified transfer functions. However, the off-line compensators cannot compensate in real-time varying dynamics of the 6-DOF EHST; so an online adaptive controller with a least-mean-squares (LMS) algorithm based on a delay compensator is employed. The proposed controller combines advantages of the off-line compensators and the online adaptive controller, which guarantees both a fast rate of convergence of the LMS algorithm and high-fidelity acceleration tracking accuracy of the 6-DOF EHST. Some experimental studies have been conducted on a 6-DOF EHST and experimental results show that acceleration tracking control performances, including the rate of convergence of the LMS algorithm and acceleration tracking accuracy, have been improved compared to a conventional three-variable controller and adaptive controllers.

1. Introduction

An electro-hydraulic shaking table (EHST) has been used in many vibration tests to evaluate structural original performances and their potential problems under actual vibrations [1–3]. The EHST generally consists of servo-valves, hydraulic actuators, a motion table, and a measurement and control system [1,4]. The purpose of the EHST is to replicate a variety of desired acceleration waveforms on a tested structure with a reasonable acceleration tracking error between the desired and actual acceleration waveforms. However, control of an EHST of six-degree-of-freedom (6-DOF) with eight hydraulic actuators to accurately track a desired acceleration signal is a challenge [5] because the 6-DOF EHST cannot normally work due to geometric effects [6], different electric parameters, and installation errors of the eight hydraulic actuators, which result in large internal coupling forces in the 6-DOF EHST. Hence, a coordinate controller is used to reduce internal coupling forces in the 6-DOF EHST. In [5,6], a general transformation framework was designed and a complete multi-axis decoupling controller was proposed for a multi-axis servo-hydraulic shaking table. An eight DOF control method was employed to implement coordinate control for a 6-DOF EHST [7]. The 6-DOF EHST can normally work with the coordinate controller and the decoupled EHST system should be excited by high-accuracy vibration drive signals [3] to obtain a high-fidelity acceleration feedback waveform. Linear feedback control, such as proportional-integral-derivative (PID) control [5], three-variables feedback control (TVC-FB) based on a combination of position, velocity, and acceleration feedback signals [8–10], and feedback linearization [11], is widely employed to improve system stability and obtain high-fidelity tracking accuracy of desired reference signals for electro-hydraulic servo systems.

However, some nonlinear and time-varying elements in electro-hydraulic servo systems caused by the dynamics of servo-valves and hydraulic actuators, such as a dead zone, friction between the rod and the bore of cylinders [12–14], and varying

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dynamics of a test specimen, deteriorate acceleration tracking performances of the 6-DOF EHST and result in difficulties to provide accurate acceleration signals to the specimen [1]. A three-variable controller (TVC) [1,3,4] is a basic controller for the EHST to improve the dynamics of servo-valves and hydraulic actuators and it consists of a three-variable feed-forward controller (TVC-FF) and a TVC-FB, which are employed to extend the frequency bandwidth of the acceleration closed-loop system and improve stability of the 6-DOF EHST by increasing natural frequencies and damping ratios of hydraulic systems, respectively. Some off-line compensators, such as the TVC, off-line iterative controllers [10], and off-line inverse model compensators (OIMCs) [3], were widely employed to compensate the dynamics of servo systems. However, the off-line compensators cannot be effectively used in nonlinear cases and this shortcoming may be remedied by using an online adaptive controller for the 6-DOF EHST, in which parameters of varying dynamics, including changes within an existing specimen during testing [5], are simultaneously adapted. Hence, some adaptive and intelligent control methods, such as adaptive inverse control [15,16], minimal control synthesis [17,18], adaptive notch filtering [1], and combined adaptive control [4,19,20], have been sought and they have attractive potentials for varying dynamics.

However, convergence performances of these above-mentioned adaptive and intelligent control algorithms, especially their rates of convergence, significantly limit control performances of these adaptive controllers for electro-hydraulic servo systems that need faster convergence. There have been numerous attempts to improve rates of convergence of these adaptive controllers using some appropriate modification methods. In [21], an approximate feed-forward inverse model controller and an adaptive controller using a velocity minimal control synthesis algorithm were proposed for a multi-axis hydraulic test rig. In [22], a frequency-domain adaptive filter with a filtered-X least-mean-squares (LMS) algorithm was employed for an adaptive inverse controller that needs faster convergence. In [23], a filtered-X LMS algorithm and an H_∞ filter were combined to improve the rate of convergence of the LMS algorithm for an electro-dynamic shaker. In [24], a novel adaptive inverse control framework with an LMS algorithm was employed to yield an accurate acceleration waveform on shaking tables.

The phase delay of the controlled closed-loop system is one of very important factors impacting on the rate of convergence of the LMS algorithm [4,25] because there is an inevitable delay in response to reference signals due to inherent dynamics of electro-hydraulic servo systems, which results in some adverse effects on the rate of convergence of the adaptive controller [26]. To solve the problem, many phase delay compensators have been presented. A feed-forward inverse model was widely employed to improve the phase delay of the controlled closed-loop system without changing its stability [4,27]. In [28], some non-minimum-phase inversion model design methods, including accurate inversion schemes and stable approximate non-minimum-phase inverse techniques, were investigated. In [29], a derivative feed-forward method was proposed for time delay compensation. In [30], an equivalent discrete transfer function was employed to analyze time delay compensation of a hydraulic actuator. In [31,32], an actuator delay compensator was designed to reduce an inevitable actuator response delay in a real-time hybrid experimental system. In [33], an adaptive controller with consideration of bounded time delays was designed for a class of multi-input multi-output nonlinear systems. An uncertain transport delay time in the transmission of an electro-hydraulic servo-valve control system was presented and a delay time variation can be effectively predicted and unconditionally removed [34].

The goal of this study is to design a combined controller to improve acceleration tracking performances of the 6-DOF EHST in the presence of varying dynamics, plant uncertainties, and modeling errors between the estimated model and the actual controlled plant. The proposed controller combines an OIMC, an improved internal model controller, and an online adaptive controller based on the LMS algorithm and a delay compensator. To develop the idea, the OIMC is employed to extend the frequency bandwidth of the acceleration closed-loop system and yield a desired measured acceleration waveform on the 6-DOF EHST, and the OIMC is designed by a zero-phase-error tracking controller using the identified acceleration closed-loop transfer function that is identified by a recursive extended least-squares (RELS) algorithm. An internal model controller is employed to minimize the effect of modeling error, and the LMS algorithm is further employed to adaptively adjust the time-domain drive signal and yield a high-fidelity acceleration response waveform on the 6-DOF EHST. A delay compensator using a linear acceleration extrapolation is designed to minimize the delay effect of hydraulic actuators and accelerate the rate of convergence of the online adaptive controller. Finally, effectiveness of the proposed controller is examined by some experiments on a 6-DOF EHST and the proposed controller yields a reasonable acceleration tracking performance.

An experimental setup of the 6-DOF EHST and a dynamic model of one hydraulic actuator are established in Section 2. Section 3 discusses the proposed controller including the coordinate controller, the TVC, the OIMC, and the internal model controller of the acceleration closed-loop system, and the delay compensator, the optimal solution, and the convergence performance of the LMS algorithm. Some simulation results and performance analysis are shown in Section 4. A series of experimental results are verified on a 6-DOF EHST to demonstrate effectiveness of the proposed controller in Section 5. Section 6 concludes main points and contributions of this study.

2. Experimental 6-DOF EHST and its dynamic model

2.1. Experimental setup

In this section, an experimental 6-DOF EHST system that is controlled by six degrees-of-freedom (six DOFs) using eight hydraulic actuators is carried out to verify effectiveness of the proposed controller in a real system. Fig. 1(a) and (b) show the experimental setup and its structural scheme, respectively, which consist of eight double-ended hydraulic actuators with a 70 mm bore and a 50 mm rod, eight flow-controlled servo-valves manufactured by Moog, Inc (G761-3004B, 38 L/min with 7 MPa pressure), sixteen spherical hinges and some relative sensors including eight linear variable differential transformers (LVDTs), eight accelerometers, and sixteen pressure transducers. The eight LVDTs are attached on the eight cylinders to measure displacements in real time; the eight accelerometers manufactured by PCB, Inc (G3711, max acceleration 2 g) are mounted on the drive rod of the eight cylinders to measure accelerations in real time, and the sixteen pressure transducers are employed to measure two chamber pressures of the eight cylinders. Main parameters of the experimental EHST system are listed in Table 1.

A block schematic of the experimental 6-DOF EHST control system is shown in Fig. 2, where a host computer serves as the user interface and enables the user to compile programs of these designed controllers and modify the control scheme and adjust parameters. A target computer serves as a real-time processing unit in which a real-time operating system (RTOS) is employed and performs real-time execution of the controller and real-time communication with A/D and D/A boards. Implementation of the digital controller and data acquisition are processed on a MATLAB/xPC target system on the target computer. Control signals are converted to analog signals by two 12-bit D/A ACL-6126 boards

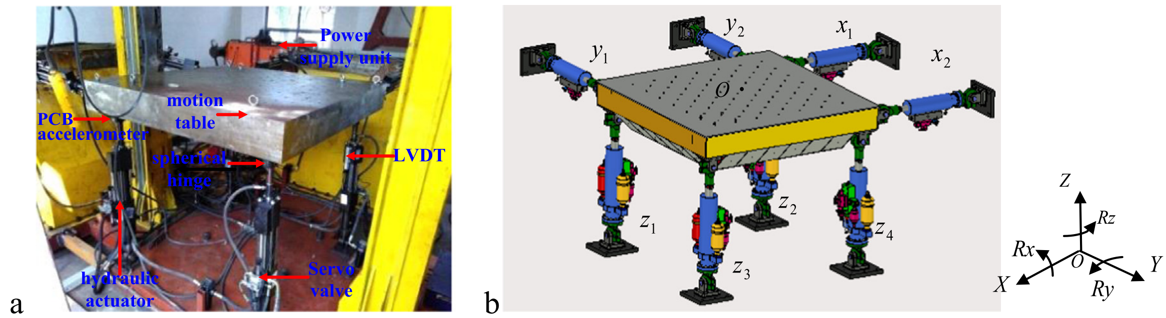

Fig. 1. Experimental system and structural scheme of the 6-DOF EHST: (a) Experimental system and (b) structural scheme.

Table 1
Main parameters of the 6-DOF EHST.

Parameters	Values
Platform size	1.5 m × 1.5 m
Platform weight	2000 kg
Payload	4000 kg
Max displacement	X, Y, and Z: ± 100 mm
Max velocity	X, Y, and Z:0.5 m/s
Max acceleration	X, Y, and Z: ± 2 g
DOF	X, Y, Z, R_x, R_y, R_z
Actuator units	8
Pressure and flow rate of the power supply unit	16 MPa, 400 L/min

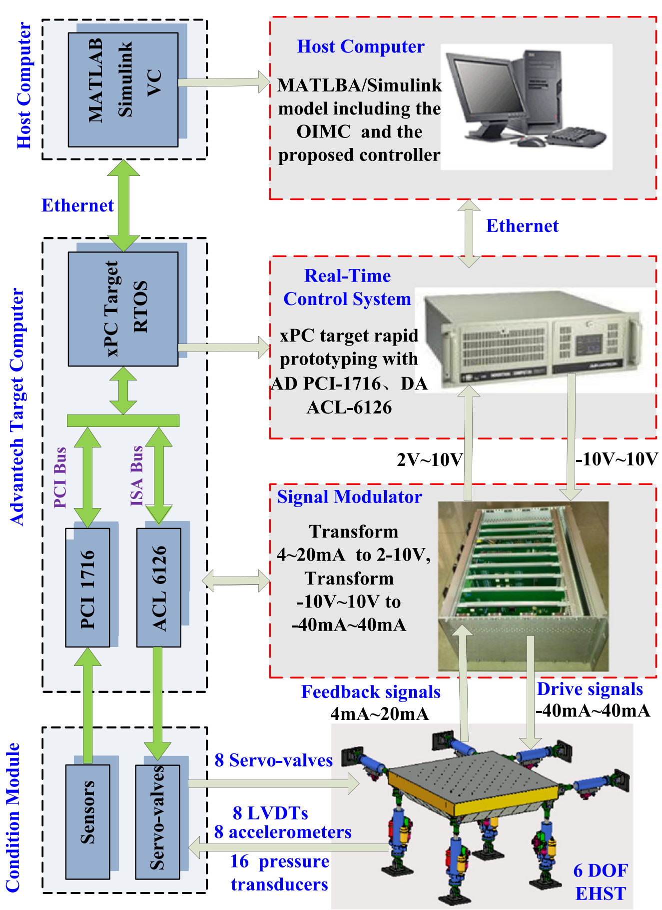

Fig. 2. Block schematic of the 6-DOF EHST control system.

and are processed by signal modulators, and are sent to servo-valves to control hydraulic actuators. Feedback measured signals in real time, including displacements, accelerations, and pressures, are collected by four 16-bit A/D PCI-1716 boards.

2.2. Dynamic model of the EHST

Referring to Fig. 1, a single hydraulic actuator configuration of the 6-DOF EHST is established because the dynamic response of each actuator is similar, which includes a platform, a servo-valve, a hydraulic cylinder and its power supply, and a displacement sensor, and their relationship is demonstrated in Fig. 3.

With combination of (A1)–(A6) in Appendix A, a dynamic model of the open-loop system can be formulated as

$$G_{op}(s) = \frac{K_0}{s\left(\frac{s^2}{\omega_{sv}^2} + \frac{2\xi_{sv}}{\omega_{sv}}s + 1\right)\left(\frac{s^2}{\omega_h^2} + \frac{2\xi_h}{\omega_h}s + 1\right)}, \tag{1}$$

where K_0 is the open-loop gain of the 6-DOF EHST, defined by $K_0 = K_{sv}K_q/A_p$, in which K_{sv} is the servo-valve gain, K_q is the linearized flow gain, and A_p is the effective area of the hydraulic actuator; ω_h and ξ_h are the hydraulic angular natural frequency and damping ratio of the actuator, respectively, which are defined by $\omega_h = \sqrt{4\beta_e A_p^2/m_t V_t}$ and $\xi_h = ((K_c + C_{tp})/A_p)\sqrt{\beta_e m_t/V_t} + (B_c/4A_p)\sqrt{V_t/\beta_e m_t}$, in which β_e is the effective bulk modulus, m_t is the total mass of the actuator piston, load and specimen, V_t is the total volume of the actuator, K_c is the flow pressure coefficient, C_{tp} is the total leakage coefficient of the actuator, including an internal leakage coefficient C_{ip} and an external leakage coefficient C_{ep}, whose relation is given by $C_{tp} = C_{ip} + C_{ep}/2$, and B_c is the viscous damping coefficient between the piston and the payload; and ω_{sv} and ξ_{sv} are the angular natural frequency and the damping coefficient of the servo-valve, respectively.

To verify dynamic characteristics of the established hydraulic model, the open-loop dynamic model is established using experimental data measured from the experimental 6-DOF EHST. The magnitude and phase characteristics of simulation and experimental results are shown in Fig. 4(a) and (b), respectively, from which it can be seen that the simulation model is able to satisfactorily match the actual experimental system. Main

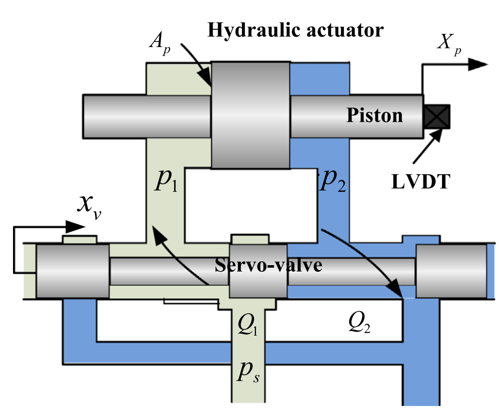

Fig. 3. Simplified structure of one actuator of the 6-DOF EHST.

simulation parameters of the hydraulic dynamic model are listed in Table 2.

3. Controller design

In this section, a servo controller is designed with consideration of the position and acceleration closed-loops, and the coordinate controller of the 6-DOF EHST, as shown in Fig. 5, where x and y are the reference and actual acceleration signals, respectively, ξ_n and ω_n are the damping ratio and the original angular frequency of the acceleration controller, respectively, and K_d is the gain of the reference signal generator. The TVC consists of the TVC-FF by adjusting three feed-forward gains, K_{ar}, K_{vr}, K_{dr}, to extend the frequency bandwidth of the acceleration closed-loop system, and the TVC-FB by adjusting three feedback gains, K_{af}, K_{vf}, K_{df}, to increase the damping ratio for improving stability of the servo controller. The reference signal generator with a second-order filter is employed to transform the reference acceleration signal to the desired displacement drive signal of the hydraulic actuator for the position closed-loop system, and two transform matrices transform between eight channel feedback signals and 6-DOF control signals.

The proposed controller is shown in Fig. 6, which includes the servo controller shown in Fig. 5, the OIMC $\hat{G}^{-1}(z)$, and the adaptive controller with the LMS algorithm. At first, the servo controller is employed to establish the position and acceleration closed-loops and the coordinate controller using two transform matrices for the 6-DOF EHST, which guarantee acceleration tracking accuracy. Secondly, the 6-DOF EHST is excited by random acceleration reference signals and input-output data are acquired to identify off-line the acceleration closed-loop system transfer function using the RELS algorithm. A stable OIMC is designed by the identified transfer function to extend the frequency bandwidth of the acceleration closed-loop and an internal model controller is designed to deal with modeling error. Finally, the adaptive controller that combines the LMS algorithm with the delay compensator is employed to cancel out varying dynamics and nonlinearities of the 6-DOF EHST for improving acceleration tracking accuracy. Hence, the proposed controller can compensate for varying dynamics and modeling error.

3.1. Coordinate control

Referring to Fig. 1(b), six DOFs of the EHST include X, Y, and Z translations, roll R_x, pitch R_y, and yaw R_z; with consideration of two redundant DOFs that are $T_h = 0.25x_1 - 0.25x_2 - 0.25y_1 + 0.25y_2$ and $T_y = 0.25z_1 - 0.25z_2 - 0.25z_3 + 0.25z_4$ [7], the DOF vector is $\mathbf{Y_{dof}} = [X,\ Y,\ R_z,\ T_h, Z, R_x,\ R_y\ , T_v]^T$. An actuator vector with eight hydraulic actuators is $\mathbf{Y_c} = [x_1,\ x_2, y_1, y_2,\ z_1,\ z_2, z_3, z_4\ ]^T$. The relationship between the DOF and actuator vectors is given by [6]

Table 2
Main parameters of the 6-DOF EHST for the simulation model.

Parameters	Description	Values
A_p	Effective area of the hydraulic actuator	1.88×10^{-3} m^2
B_c	Viscous damping coefficient	25,000 Ns/m
C_{ep}	External leakage coefficient	4.6×10^{-17} m^3/(s/Pa)
C_{ip}	Internal leakage coefficient	4.6×10^{-17} m^3/(s/Pa)
K_q	Linearized flow gain	0.00145 (m^3/s)/V
K_c	Flow pressure coefficient	2×10^{-12} m^3/(s Pa)
K_{sv}	Servo-valve gain	4 m^3/s/A
m_t	Total mass	500 kg
V_t	Total volume of the actuator	0.96×10^{-3} m^3
ω_h	Angular natural frequency of actuator	32 Hz
ω_{sv}	Angular natural frequency of servo-valve	100 Hz
ξ_h	Damping ratio of the actuator	0.3
ξ_{sv}	Damping coefficient of the servo-valve	0.8
β_e	Effective bulk modulus	6.9×10^8 Pa

$$\mathbf{A}\mathbf{Y}_{\mathrm{dof}} = \mathbf{Y_c}, \tag{2}$$

where $\mathbf{A}$ is a transform matrix from eight hydraulic actuators to eight DOFs with two redundant DOFs. One has

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0.25 & -0.25 & 0.25 & -0.25 & 0 & 0 & 0 & 0 \\ 0.25 & -0.25 & -0.25 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.25 & 0.25 & 0.25 \\ 0 & 0 & 0 & 0 & 0.25 & 0.25 & -0.25 & -0.25 \\ 0 & 0 & 0 & 0 & 0.25 & -0.25 & 0.25 & -0.25 \\ 0 & 0 & 0 & 0 & 0.25 & -0.25 & -0.25 & 0.25 \end{bmatrix}, \tag{3}$$

and $\mathbf{A}^{-1}$ is a transform matrix from eight DOFs to eight hydraulic actuators, which is given by

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}. \tag{4}$$

3.2. Three-variable controller

The TVC is a conventional method for the EHST, which is shown in Fig. 7, where $x_d(k)$ and $y_d(k)$ are the reference and feedback position signals, respectively. The TVC consists of the TVC-FF and the TVC-FB, which are employed to extend the frequency bandwidth of the acceleration closed-loop system and improve stability of the servo controller by increasing the angular natural frequency

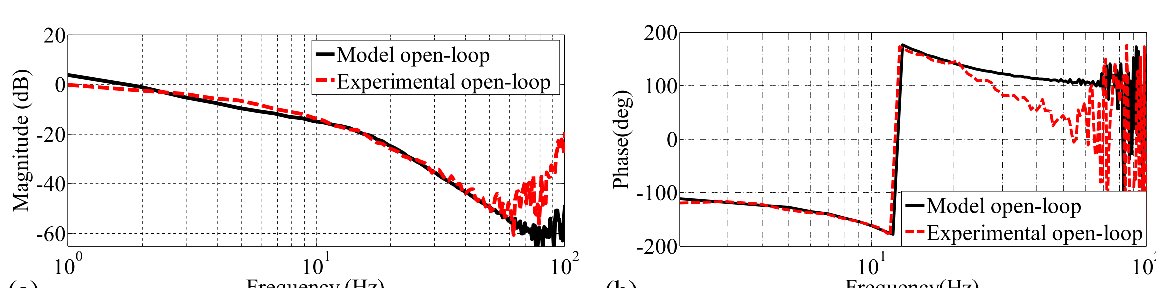

Fig. 4. Open-loop frequency response characteristics of simulation and experimental models for one actuator of the 6-DOF EHST: (a) magnitude and (b) phase.

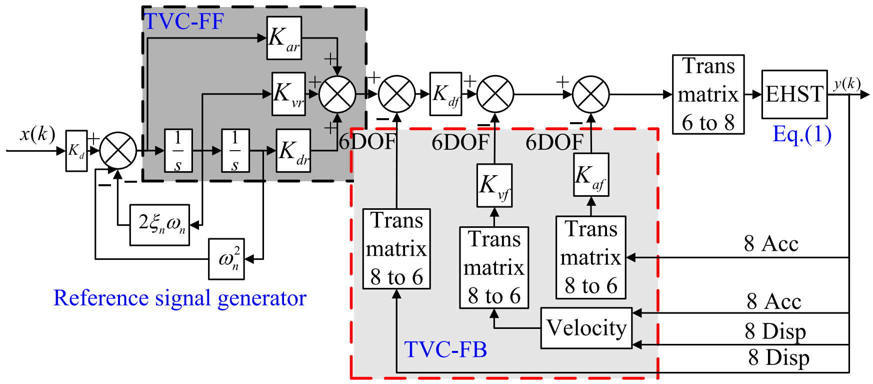

Fig. 5. The servo controller for the 6-DOF EHST.

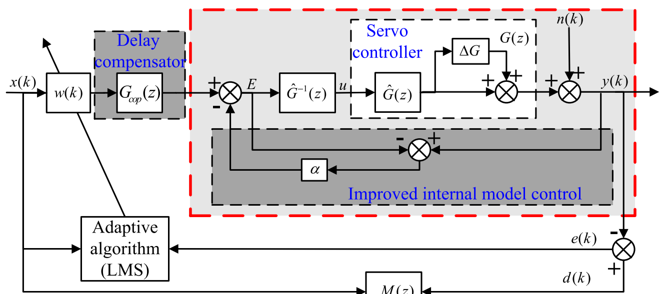

Fig. 6. The proposed controller for the 6-DOF EHST.

and the damping ratio of the hydraulic system, respectively. A reference signal generator with $K_d/(s^2+2\xi_n\omega_n s+\omega_n^2)$ is employed to convert the acceleration signal to the position drive signal because the control mode of the 6-DOF EHST is acceleration control; however, the design method of the TVC is the position closed-loop.

Referring to Fig. 7, a position open-loop transfer function with the acceleration and velocity feedback signals is expressed by

$$G_{op}(s)=\frac{K_0K_{df}}{s\left(\frac{s^2}{\omega_{sv}^2}+\frac{2\xi_{sv}}{\omega_{sv}}s+1\right)\left(\frac{s^2}{\omega_h^2}+\frac{2\xi_h}{\omega_h}s+1\right)+K_0\left(K_{af}s^2+K_{vf}s\right)}=\frac{K_0K_{df}/(1+K_0K_{vf})}{s\left(\frac{s^2}{\omega_0^2}+\frac{2\xi_0}{\omega_0}s+1\right)\left(\frac{s^2}{\omega_{sv}^2}+\frac{2\xi_{sv}}{\omega_{sv}}s+1\right)},\tag{5}$$

where ω_0 and ξ_0 are the angular natural frequency and the damping ratio of the hydraulic actuator with the acceleration and velocity feedback signals, respectively, which are defined by $\omega_0=\omega_h\sqrt{1+K_0K_{vf}}$ and $\xi_0=(\xi_h+K_{af}K_0)/\sqrt{1+K_0K_{vf}}$. It can be noticed that ω_h and ξ_h are improved by the acceleration and velocity feedback signals. With consideration of dynamic characteristics of the servo-valve, the position closed-loop transfer function with the TVC-FB is expressed by

$$G_{cp}(s)=\frac{K_0K_{df}}{\frac{s^5}{\omega_{sv}^2\omega_h^2}+\left(\frac{2\xi_h}{\omega_{sv}^2\omega_h}+\frac{2\xi_{sv}}{\omega_{sv}\omega_h^2}\right)s^4+\left(\frac{1}{\omega_{sv}^2}+\frac{2\xi_h\xi_{sv}}{\omega_{sv}\omega_h}+\frac{1}{\omega_h^2}\right)s^3+\left(\frac{2\xi_{sv}}{\omega_{sv}}+\frac{2\xi_h}{\omega_h}+K_{af}K_0\right)s^2+\left(1+K_0K_{vf}\right)s+K_0K_{df}+1}.\tag{6}$$

Define the desired position closed-loop system transfer function as [3]

$$W(s)=\frac{1}{\left(\frac{s}{\omega_r}+1\right)\left(\frac{s^2}{\omega_{nc}^2}+\frac{2\xi_{nc}}{\omega_{nc}}s+1\right)\left(\frac{s^2}{\omega_{sv}^2}+\frac{2\xi_{sv}}{\omega_{sv}}s+1\right)},\tag{7}$$

where ω_r is the frequency bandwidth of the desired position closed-loop, $\omega_{nc}=1.05\omega_h\text{~}1.2\omega_h$, and $\xi_{nc}\approx0.7$. By making the closed-loop transfer function to be consistent with the desired closed-loop, three feedback gains K_{af}, K_{vf}, K_{df} of the TVC-FB are given by

$$\begin{cases}K_{df}=\dfrac{\omega_r\omega_{nc}^2}{K_0\omega_h^2}\\ K_{vf}=K_{df}(\dfrac{2\xi_{nc}}{\omega_{nc}}+\dfrac{1}{\omega_r})-\dfrac{1}{K_0}\\ K_{af}=K_{df}(\dfrac{2\xi_{nc}}{\omega_r\omega_{nc}}+\dfrac{1}{\omega_{nc}^2})-\dfrac{2\xi_h}{K_0\omega_h}\end{cases}.\tag{8}$$

In order to eliminate poles that are close to the imaginary axis of the position closed-loop to extend the frequency bandwidth of the EHST system, referring to Fig. 7, the TVC-FF is designed by

$$B(s)=K_{dr}(1+\frac{K_{dr}}{K_{dr}}s+\frac{K_{ar}}{K_{dr}}s^2)=\left(\frac{s^2}{\omega_{nc}^2}+\frac{2\xi_{nc}}{\omega_{nc}}s+1\right).\tag{9}$$

To ensure the position closed-loop system gain to be unchanged, set $K_{dr}=1$; three feed-forward gains K_{ar}, K_{vr}, K_{dr} of the TVC-FF are given by

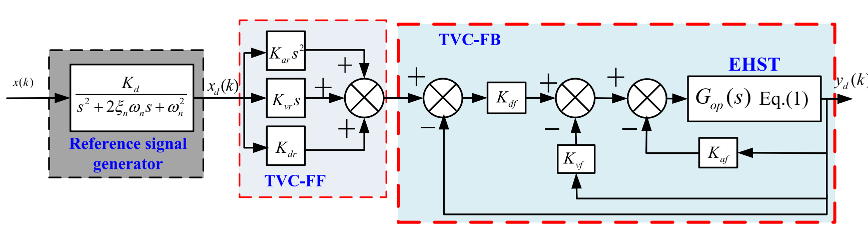

Fig. 7. The TVC for the 6-DOF EHST.

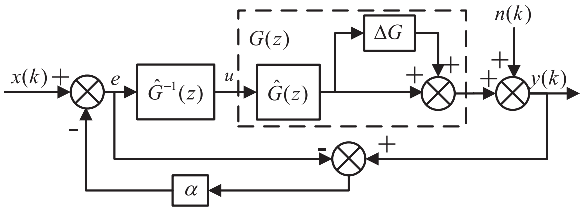

Fig. 8. Block diagram of the improved internal model controller.

$$\begin{cases} K_{dr} = 1 \\ K_{vr} = \dfrac{2\xi_{nc}}{\omega_{nc}} . \\ K_{ar} = \dfrac{1}{\omega_{nc}^2} \end{cases} \tag{10}$$

3.3. Off-line inverse model compensator

In order to design the OIMC, the acceleration closed-loop transfer function is identified using the RELS algorithm that is shown in Appendix B. The identified acceleration closed-loop transfer function can be written as

$$\hat{G}(z) = \frac{\hat{B}(z)}{\hat{A}(z)} = \frac{\hat{B}_a(z)\hat{B}_u(z)}{\hat{A}(z)}, \tag{11}$$

where $\hat{A}(z)$ is the denominator polynomial that contains all the acceleration closed-loop system poles, which defined by $\hat{A}(z) = z^n + a_1 z^{(n-1)} + \cdots + a_n$, $\hat{B}(z)$ is the numerator polynomial and is decomposed into $\hat{B}_a(z)$ and $\hat{B}_u(z)$, where $\hat{B}_a(z)$ contains all plant minimum phase zeros and $\hat{B}_u(z)$ contains all non-minimum phase zeros, which are defined by $\hat{B}_a(z) = b_{ap}z^p + b_{a(p-1)}z^{(p-1)} + \cdots + b_{a0}$ and $\hat{B}_u(z) = b_{uq}z^q + b_{u(q-1)}z^{(q-1)} + \cdots + b_{u0}$. The stable OIMC of the identified transfer function can be formulated by

$$\hat{G}^{-1}(z) = \frac{\hat{B}_u(z^{-1})\hat{A}(z)}{\hat{B}_a(z)B_u^2(1)z^{\gamma}}, \tag{12}$$

For physical realizability of $\hat{G}^{-1}(z)$, z^{γ} is introduced into (12). Combining (11) and (12) yields

$$H(z) = \hat{G}(z)\hat{G}^{-1}(z) = \frac{\hat{B}_u(z)\hat{B}_u(z^{-1})}{z^{\gamma}\hat{B}_u^2(1)}, \tag{13}$$

Substituting z in (13) by $e^{j\omega T}$ and noting that $\hat{B}_u(e^{jwT})$ and $\hat{B}_u(e^{-jwT})$ are complex conjugates to each other, one has

$$\angle H(e^{j\omega T}) = -s\omega T, \tag{14}$$

$$\left|H(e^{j\omega T})\right|^2 = \mathrm{Re}^2\left[H(e^{j\omega T})\right] + \mathrm{Im}^2\left[H(e^{j\omega T})\right]. \tag{15}$$

If ωT is close to 0 in (15), $\left|H_i(e^{j\omega T})\right|$ is close to 1.

3.4. Delay compensator

The rate of convergence of the online adaptive controller is limited in the 6-DOF EHST during real-time testing due to the hydraulic actuator delay. In [28], an actuator delay compensator was proposed using a linear acceleration extrapolation method and it was simplified in this work to compensate the hydraulic actuator delay for improving acceleration tracking performances. With consideration of the time step Δt and the hydraulic actuator delay τ, the linearly varying acceleration assumption can be expressed by [28]

$$\begin{aligned}\ddot{x}(t+\Delta t) &= \ddot{y}(t) + \frac{\Delta t + \tau}{\Delta t}\cdot[\ddot{y}(t) - \eta\cdot\ddot{y}(t-\Delta t)] = (2 + \frac{\tau}{\Delta t})\cdot\ddot{y}(t) \\ &\quad - \eta\cdot\frac{\tau}{\Delta t}\cdot\ddot{y}(t-\Delta t),\end{aligned} \tag{16}$$

where η is an adjustable parameter. Employing the discrete z-transform in (16), one has

$$X(z) = Y(z)\cdot[(2 + \frac{\tau}{\Delta t})z - \eta\cdot(1 + \frac{\tau}{\Delta t})]\cdot\frac{1}{z^2}, \tag{17}$$

Referring to (17), the hydraulic actuator delay compensator can be given by

$$G_{cop}(z) = \frac{X(z)}{Y(z)} = [(2 + \frac{\tau}{\Delta t})z - \eta\cdot(1 + \frac{\tau}{\Delta t})]\cdot\frac{1}{z^2}. \tag{18}$$

3.5. Internal model controller

In order to further improve acceleration tracking performances of the 6-DOF EHST, an improved internal model controller is introduced, as shown in Fig. 8, where $G(z)$ is an actual acceleration closed-loop transfer function, $\hat{G}(z)$ is an estimated model of $G(z)$, $\hat{G}^{-1}(z)$ is an inverse model of $\hat{G}(z)$, and ΔG depicts a modeling error between the estimated model $\hat{G}(z)$ and the actual plant $G(z)$, which is caused by noise, unmodeled dynamics, and nonlinearities [35]. An improved internal model controller is employed by tuning its gain $\alpha(0 < \alpha < 1)$ to minimize the detrimental effect of the modeling error ΔG. The acceleration output response signal $y(k)$ of the EHST is disturbed by uncertain noise $n(k)$. With combination of Fig. 6, the desired signal $d(k)$ is generated by a reference model $M(z)$, which is driven by the reference input signal $x(k)$. An acceleration tracking error signal $e(k) = d(k) - y(k)$ and the input signal $x(k)$ are drawn into the LMS algorithm to update the adaptive filter weight coefficient $w(k)$, as shown in Fig. 6. The controlled acceleration closed-loop system becomes a nearly unit gain and zero phase shift system across the entire operating frequency range by designing $\hat{G}^{-1}(z)$ and adjusting α.

According to Fig. 8, by neglecting the internal model controller, the output response signal $y(k)$ can be formulated as

$$y(k) = \hat{G}^{-1}(z)\hat{G}(z)(1+\Delta G)x(k) + n(k) = (1+\Delta G)x(k) + n(k). \tag{19}$$

With consideration of the improved internal model controller, the acceleration output response signal $y(k)$ can be expressed by

$$y(k) = \frac{(1+\Delta G)}{1+\alpha\Delta G}x(k) + \frac{1-\alpha}{1+\alpha\Delta G}n(k). \tag{20}$$

Referring to (19) and (20), the acceleration tracking error between the reference signal and the output response signal can be derived by

$$Err(k) = (1+\Delta G)x(k) + n(k) - x(k) = x(k)\Delta G + n(k), \tag{21}$$

$$\begin{aligned}Err^*(k) &= \frac{(1+\Delta G)}{1+\alpha\Delta G}x(k) + \frac{1-\alpha}{1+\alpha\Delta G}n(k) - x(k) \\ &= \frac{1-\alpha}{1+\alpha\Delta G}(x(k)\Delta G + n(k)),\end{aligned} \tag{22}$$

respectively. As can be seen from (21) and (22), the acceleration tracking error decreases in (22) due to the improved internal model controller. Referring to Figs. 6 and 8, the input vector $\mathbf{x}(k)$ and adaptive filter weight coefficients $\mathbf{w}(k)$ are defined by $\mathbf{x}(k) = [x(k-1), \cdots, x(k-M)]^T$ and $\mathbf{w}(k) = [w(k-1), \cdots, w(k-M)]$, where M is the adaptive filter length. With combination of the delay compensator in (20) and the adaptive controller, the acceleration output signal $y(k)$ can be expressed by

$$y(k)=\frac{(1+\Delta G)G_{cop}(z)w(k)}{1+\alpha\Delta G}x(k)+\frac{1-\alpha}{1+\alpha\Delta G}n(k). \tag{23}$$

With consideration of (23), the acceleration tracking error can be written as

$$e(k)=M(z)x(k)-y(k)=[z^{-L}-\frac{(1+\Delta G)G_{cop}(z)w(k)}{1+\alpha\Delta G}]x(k)$$
$$-\frac{1-\alpha}{1+\alpha\Delta G}n(k), \tag{24}$$

where $M(z)$ is the reference model and it is chosen as the response delay of the plant, which appropriately requires z^{-L} to be of the order of half the length of the adaptive filter.

3.6. Optimal solution

The objective of the adaptive LMS algorithm is to iteratively find an optimal controller to minimize the expected root-mean-square error. A cost function $J(k)$ can be defined by [36]

$$J(k)=E[e^2(k)], \tag{25}$$

where $E[\cdot]$ denotes the statistical expectation. By neglecting the correlation between $x(k)$ and $n(k)$ and substituting (24) into (25), $J(k)$ can be written as

$$J(k)=E[(z^{-L}-\frac{(1+\Delta G)G_{cop}(z)w(k)}{1+\alpha\Delta G})^2x(k)x(k)^T]$$
$$+E[(\frac{1-\alpha}{1+\alpha\Delta G})^2n^2(k)]. \tag{26}$$

The minimum value $J_{\min}$ of $J(k)$ in (26) can be derived by setting its partial gradient with respect to $w(k)$ to be zero because $J(k)$ is a convex function of $w(k)$. One has

$$\frac{\partial J}{\partial w(k)}=-2E[\frac{(1+\Delta G)}{1+\alpha\Delta G}z^{-L}]R+2E[(\frac{(1+\Delta G)}{1+\alpha\Delta G})^2]G_{cop}(z)w(k)R$$
$$=0, \tag{27}$$

where $\mathbf{R}$ is the auto-correlation matrix defined by $\mathbf{R}=E[x(k)x(k)^T]$, which is the positive definite or positive semi-definite because its root-mean-square error is non-negative. The optimum solution of the proposed controller is given by

$$w(k)_{\text{opt}}=\frac{1+\alpha\Delta G}{(1+\Delta G)G_{cop}(z)}z^{-L}. \tag{28}$$

By substituting (28) into (24), the minimum value $J_{\min}$ of the cost function can be obtained

$$J_{\min}=E[e^2(k)]|_{w(k)=w(k)_{\text{opt}}}=E[(\frac{1-\alpha}{1+\alpha\Delta G})^2n^2(k)]. \tag{29}$$

With consideration of different conditions, different optimum solutions are given by

$$w(k)_{\text{opt}}=\begin{cases}\dfrac{1+\alpha\Delta G}{(1+\Delta G)G_{cop}(z)}z^{-L} & \alpha\neq 0\\[2ex] \dfrac{1}{(1+\Delta G)G_{cop}(z)}z^{-L} & \alpha=0\end{cases}. \tag{30}$$

Ideally, a tuned plant would have a transfer function between the command and feedback signals characterized by a unit gain and zero phase shifts across the entire operating frequency range. Hence, a unit value minus two different optimum solutions in (30) yields

$$\left|1-\frac{1}{(1+\Delta G)G_{cop}(z)}\right|\geq\left|1-\frac{1+\alpha\Delta G}{(1+\Delta G)G_{cop}(z)}\right|. \tag{31}$$

Note from (31) that the proposed controller can yield a better optimum solution compared to other controllers. Therefore, effectiveness of the proposed controller is verified.

3.7. Convergence analysis

The weight update of the LMS algorithm is given by [37]. Defining $w^*(k)=G_{cop}(z)w(k)$, one has

$$w^*(k+1)=w^*(k)+\rho e(k)x(k), \tag{32}$$

$$\rho=\frac{\mu}{\frac{\beta}{\lambda+\sum_{i=k-l+1}^{k}e^2(k)}+(1-\beta)\|x(k)\|^2}, \tag{33}$$

where β ($0<\beta<1$) and λ ($0<\lambda<1$) are small positive constants, l ($0<l<k$) is the length of a data window, μ is a step-size parameter, β is used to finely tune the step-size μ, and λ is employed to prevent the denominator from being zero. The step-size μ satisfies the following equation by means of Appendix C.

$$0<\mu<\frac{\frac{\beta}{\lambda}+(1-\beta)R}{\lambda_{\max}}\frac{1+\alpha\Delta G}{(1+\Delta G)}\leq[\frac{\beta}{\lambda\lambda_{\max}}+N(1-\beta)]\frac{1+\alpha\Delta G}{(1+\Delta G)}, \tag{34}$$

where $\lambda_{\max}$ is the largest eigenvalue of $\mathbf{R}$.

Weight update of the Filtered-X LMS and ε-Filtered LMS algorithms are given by[37]

$$w(k+1)=w(k)+\rho e(k)[\hat{G}(z)x(k)], \tag{35}$$

$$w(k+1)=w(k)+\rho[\hat{G}^{-1}(z)e(k)]x(k). \tag{36}$$

The optimum solution and the maximum step of three adaptive algorithms including the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller, are compared in Table 3. It can be observed that convergence properties of the proposed controller are improved.

4. Simulation studies

To verify the performance of the proposed controller, simulations are performed on the established simulation model shown in (1). Mean-square errors of the acceleration tracking performance excited by a sine signal with the amplitude of 0.2 g and the frequency of 40 Hz, and a random signal with the amplitude of 1 g and the frequency range of 2–50 Hz are shown in Fig. 9 using the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller. It can be clearly observed that the proposed controller have the fastest rate of convergence and the highest accuracy compared to the Filtered-X LMS and ε-Filtered LMS algorithms.

Table 3
Performance comparison of the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller, where $\gamma=\beta/\lambda\lambda_{\max}+N(1-\beta)$.

Algorithms	Optimum solution	Maximum step
Filtered-X LMS	$z^{-L}/G(z)(1+\Delta G)$	$0<\mu<\gamma/G(z)G(z)(1+\Delta G)$
ε-Filtered LMS	$z^{-L}/(1+\Delta G)$	$0<\mu<\gamma/(1+\Delta G)$
Proposed	$(1+\alpha\Delta G)z^{-L}/(1+\Delta G)G_{cop}(z)$	$0<\mu<\gamma(1+\alpha\Delta G)/(1+\Delta G)$

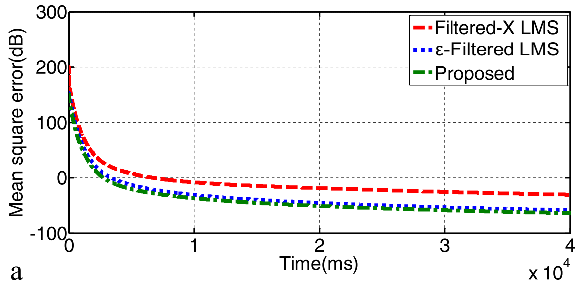

Fig. 9. Simulation results of mean-square errors with the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller using different acceleration reference signals: (a) 40 Hz sine and (b) 2–50 Hz random.

5. Experimental studies

5.1. Experimental results of the TVC

The servo controller shown in Fig. 5 is first used to implement the acceleration closed-loop control for the 6-DOF EHST. A series of experimental tests of the position and acceleration tracking performances have been performed using step command signals with 0.02 m position and 1 g acceleration in the Z direction of the 6-DOF EHST because the dynamic response of each DOF in the 6-DOF EHST is similar. Experimental results of the position and acceleration step output responses are shown in Fig. 10. The TVC used has been manually tuned and high feedback and feed-forward gains have been tuned to maximize control performances. The tuned parameters of the TVC are listed in Table 4. It can be seen that the TVC-FB increases the damping ratio by reducing the oscillation shown in Fig. 10(a) and the TVC-FF extends the acceleration closed-loop frequency bandwidth of the EHST shown in Fig. 10(b). These step performances work well for six DOFs. Some random acceleration experiments with the amplitude of 1 g and the frequency range of 2–80 Hz are then applied to six DOFs and experiments with the servo controller are performed to demonstrate effectiveness of designed transform matrices shown in (3) and (4) using tuned parameters in Table 4. Experimental results for six DOFs are shown in Fig. 11, from which it can be noticed that some satisfactory experimental results (translational and rotational magnitude frequency bandwidths are 15 Hz and 60 Hz within − 3 dB, respectively, and their phase frequency bandwidths are about 20 Hz within − 90°) are obtained by tuning the TVC-FB and the TVC-FF. Hence, use of the TVC can yield a better acceleration tracking performance.

5.2. Experimental results of the OIMC

The 6-DOF EHST is a parallel over-constrained system and the dynamic output response of each DOF is similar. Hence, an acceleration reference random signal with the amplitude of 1 g and the frequency range of 2–80 Hz is first conducted to identify frequency response characteristics of the Z-direction acceleration closed-loop system. A 15 order-discrete acceleration transfer function is identified as

Table 4
Tuned experimental parameters for six DOFs.

DOF	K_{ar}	K_{vr}	K_{dr}	K_{df}	K_{vf}	K_{af}
X	0.028	0.5	0.85	15	0.005	0.04
Y	0.05	2	1	15	0.005	0.06
Z	0.007	1.5	1	15	0.005	0.1
R_x	0.008	2	1	15	0.005	0.05
R_y	0.005	1.2	1	15	0.005	0.001
R_z	0.009	1.5	1	15	0.005	0.06

$$\hat{G}(z)=\frac{\begin{aligned}&-0.4368z^{15}+4.002z^{14}-16.08z^{13}+36.94z^{12}-52.11z^{11}+43.25z^{10}\\&-14.37z^{9}-6.368z^{8}-3.277z^{7}+37.18z^{6}-68.03z^{5}+71.95z^{4}-49.61z^{3}\\&+22.01z^{2}-5.715z+0.6609\end{aligned}}{\begin{aligned}&z^{15}-5.951z^{14}+13.72z^{13}-12.66z^{12}-3.637z^{11}+16.91z^{10}-7.119z^{9}\\&-12.44z^{8}+13.38z^{7}+4.731z^{6}-17.34z^{5}+12.47z^{4}-1.65z^{3}-2.782z^{2}\\&+1.688z-0.3134\end{aligned}} \tag{37}$$

To obtain poles and zeros of the identified model, the identified discrete acceleration transfer function can be expressed by

$$\hat{G}(z)=\frac{\begin{aligned}&-0.43683(z-0.9946)(z-0.6331)(z-2.001)(z^2-1.953z+0.9591)\\&(z^2-1.845z+0.9648)(z^2-1.735z+0.8952)(z^2+1.912z+1.13)\\&(z^2-1.887z+1.407)(z^2-0.02558z+0.9114)\end{aligned}}{\begin{aligned}&(z+1)(z+0.5417)(z-0.9893)(z^2-1.967z+0.9708)\\&(z^2-1.797z+0.8705)(z^2-1.847z+0.9683)(z^2-1.777z+0.9395)\\&(z^2+1.597z+0.9976)(z^2-0.7132z+0.7628)\end{aligned}} \tag{38}$$

A stable OIMC can be designed by

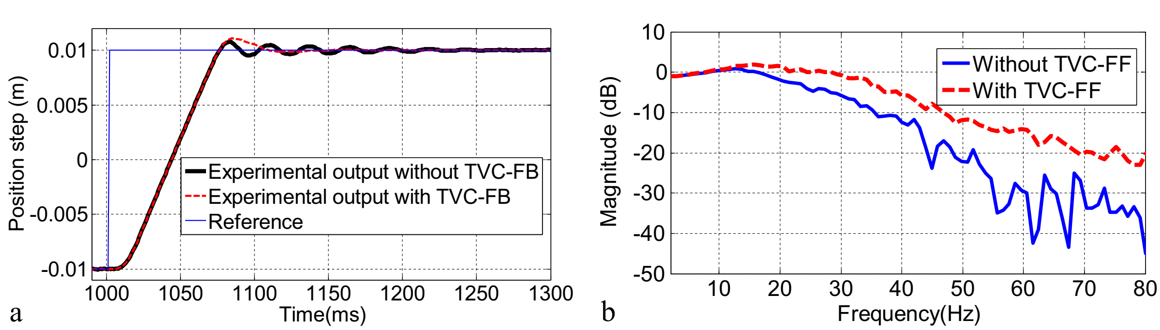

Fig. 10. Experimental results in the Z direction with the TVC: (a) the position and (b) acceleration closed-loops.

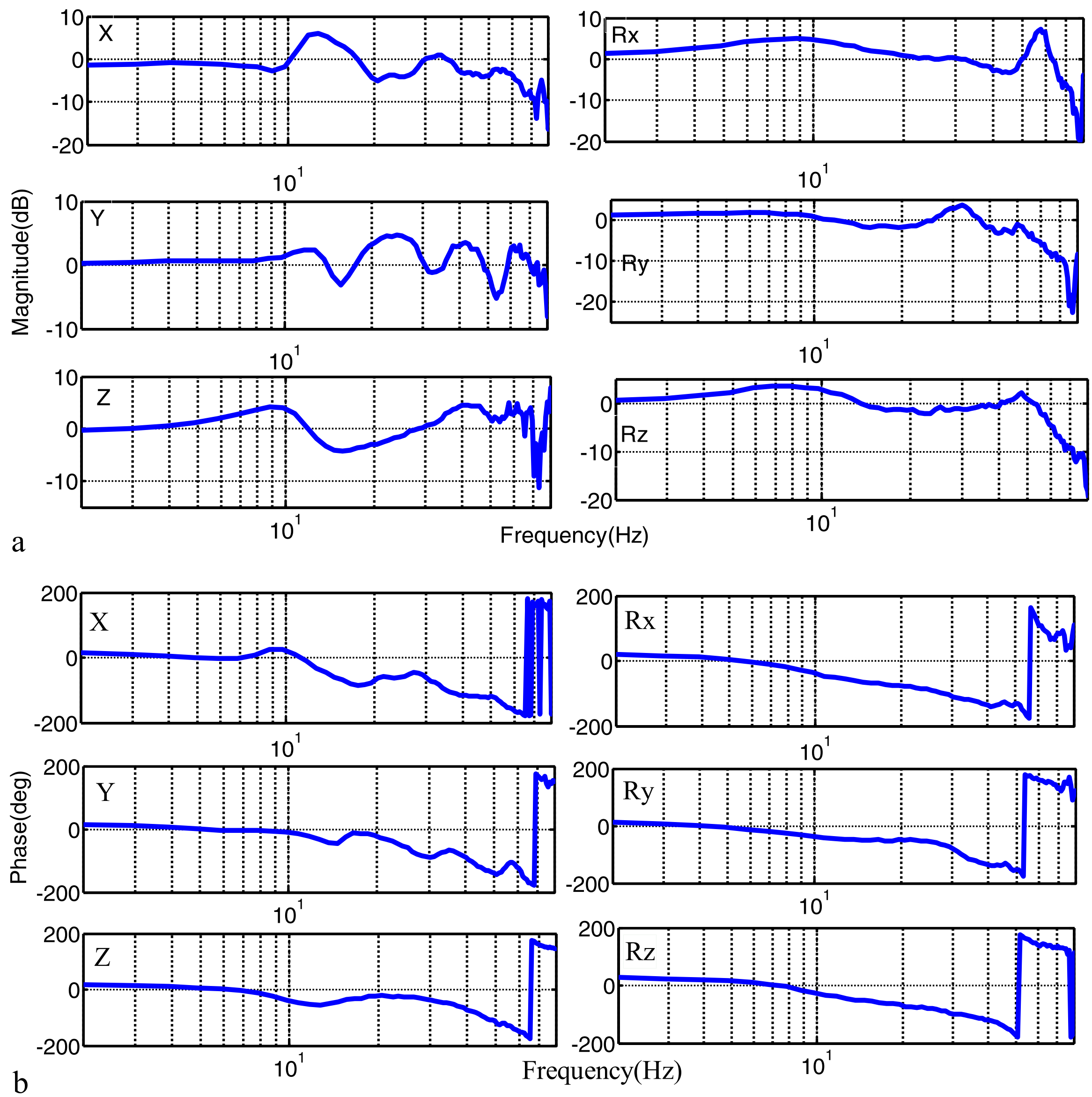

Fig. 11. Frequency response characteristics of experimental results with the TVC using acceleration random command signals with the frequency range of 2–80 Hz for the 6-DOF EHST: (a) magnitude and (b) phase.

$$\hat{G}^{-1}(z) = \frac{\begin{array}{l}(z+1)(z+0.5417)(z-0.9893)(z^2-1.967z+0.9708)\\(z^2-1.797z+0.8705)(z^2-1.847z+0.9683)(z^2-1.777z+0.9395)\\(z^2+1.597z+0.9976)(z^2-0.7132z+0.7628)\end{array}}{\begin{array}{l}-0.43683(z-0.9946)(z-0.6331)(2.001z-1)(z^2-1.953z+0.9591)\\(z^2-1.845z+0.9648)(z^2-1.735z+0.8952)(1.13z^2+1.912z+1)\\(1.407z^2-1.887z+1)(z^2-0.02558z+0.9114)\end{array}} \tag{39}$$

Comparison results of frequency response characteristics for the experimental, identified, and designed inverse models are shown in Fig. 12. Two results indicate that the experimental and identified models have a good match and the designed inverse transfer function can effectively compensate the actual acceleration closed-loop system in their frequency ranges.

5.3. *Experimental results of the proposed controller*

The proposed controller is developed by incorporating the LMS algorithm and the OIMC shown in (39), the delay compensator expressed by $(56z-40)/z^2$, and the gain of the internal model controller $\alpha = 0.2$. In order to verify effectiveness of the proposed controller when the 6-DOF EHST operates in real working conditions, a sine acceleration reference signal with the amplitude of 0.2 g and the frequency of 40 Hz is applied in the Z-direction. The experimental result with the proposed controller is compared to those with the Filtered-X LMS and ε-Filtered LMS algorithms. The adaptive tuning process of three controllers and their mean-square errors are shown in Fig. 13. The proposed controller yields a significant fast rate of convergence compared to those from the Filtered-X LMS and ε-Filtered LMS algorithms; the convergence time is reduced from 35 s with the ε-Filtered LMS algorithm to 10 s with the Filtered-X LMS algorithm, and 2 s with the proposed controller, as shown in Fig. 13(a-c). In addition, mean-square errors shown in Fig. 13(d) using the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller are − 5 dB, − 10 dB and − 20 dB, respectively. Detailed measured acceleration sine waveforms of Fig. 13 with the TVC, the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller are shown in Fig. 14 (a-d).

A random acceleration reference waveform with the amplitude of 1 g and the frequency range of 2–50 Hz in the Z-direction is

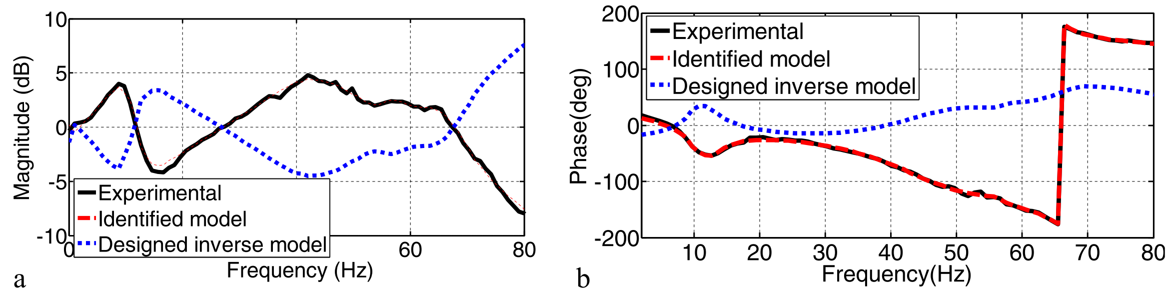

Fig. 12. Frequency response characteristics of the experimental, identified, and designed inverse models in the Z direction: (a) magnitude and (b) phase.

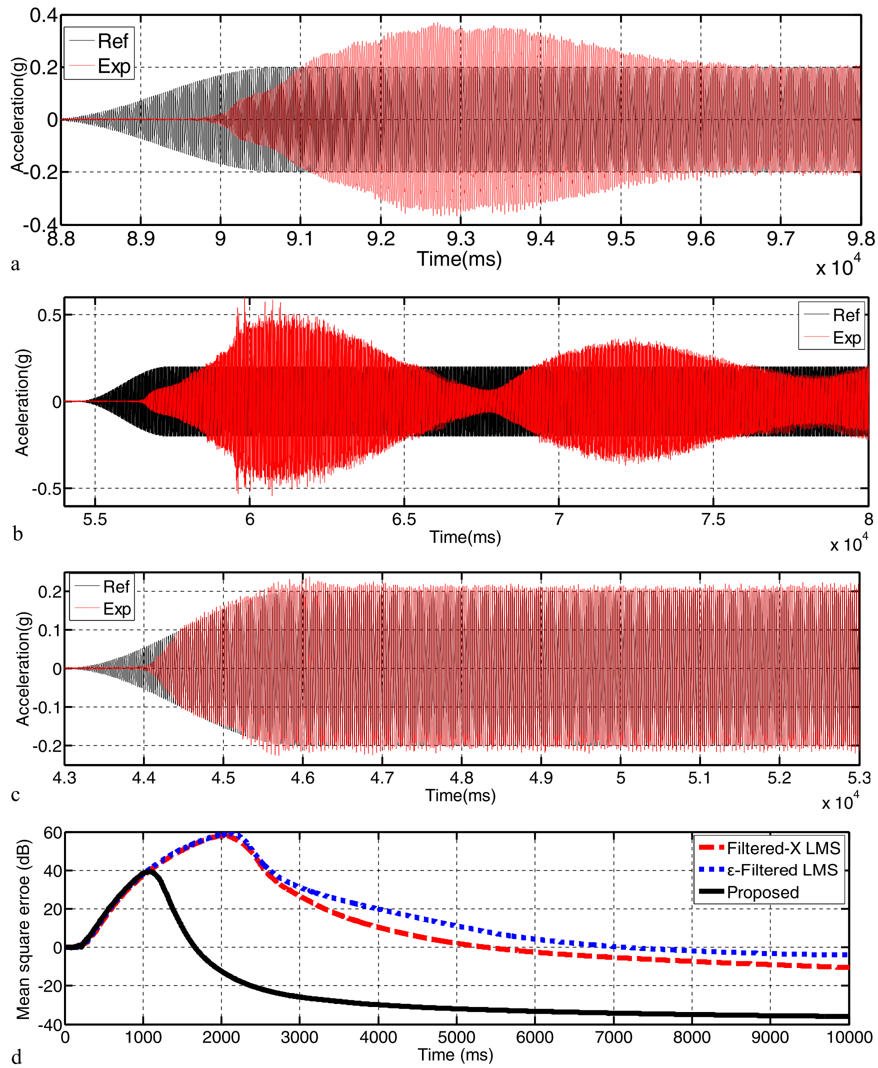

Fig. 13. Experimental results using a 40 Hz acceleration sine signal: (a) the Filtered-X LMS algorithm, (b) the ε-Filtered LMS algorithm, and (c) the proposed controller, and (d) their mean-square errors.

employed to verify control performances of the proposed controller. Fig. 15 shows the adaptive online tuning process with the random input signal. Fig. 15(a-c) show that convergence times with the same random reference excitation signal using the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller are 20 s, 7 s, and 2 s, respectively. Mean-square errors shown in Fig. 15(d) are − 10 dB, − 25 dB, and − 39 dB using the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller, respectively. Time domain waveforms of acceleration random signals with the TVC, the Filtered-X LMS and ε-Filtered LMS algorithms, and the proposed controller are shown in Fig. 16(a–d).

It can be obviously seen from Figs. 13–16 that acceleration tracking performances using the proposed controller are the best and mean-square errors using the proposed controller are always the smallest whether with the 40 Hz sine excitation signal or the 2–50 Hz random excitation signal. In conclusion, the proposed controller yields a smaller tracking error and a faster rate of convergence than the Filtered-X LMS and ε-Filtered LMS algorithms.

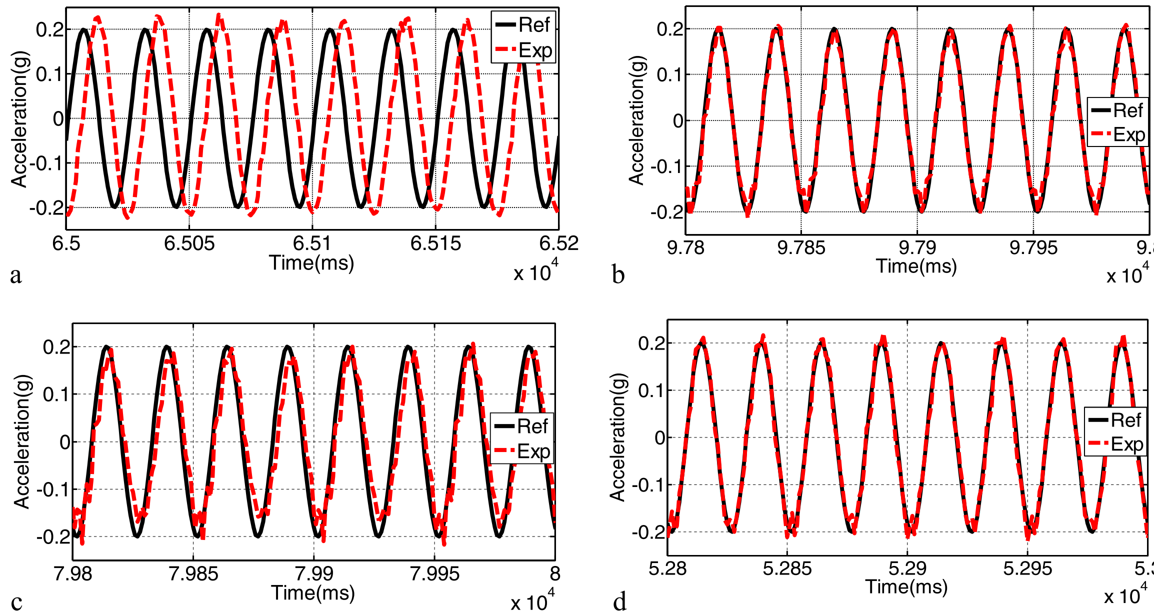

Fig. 14. Experimental results using a 40 Hz sine signal: (a) the TVC, (b) the Filtered-X LMS algorithm, (c) the ε-Filtered LMS algorithm, and (d) the proposed controller.

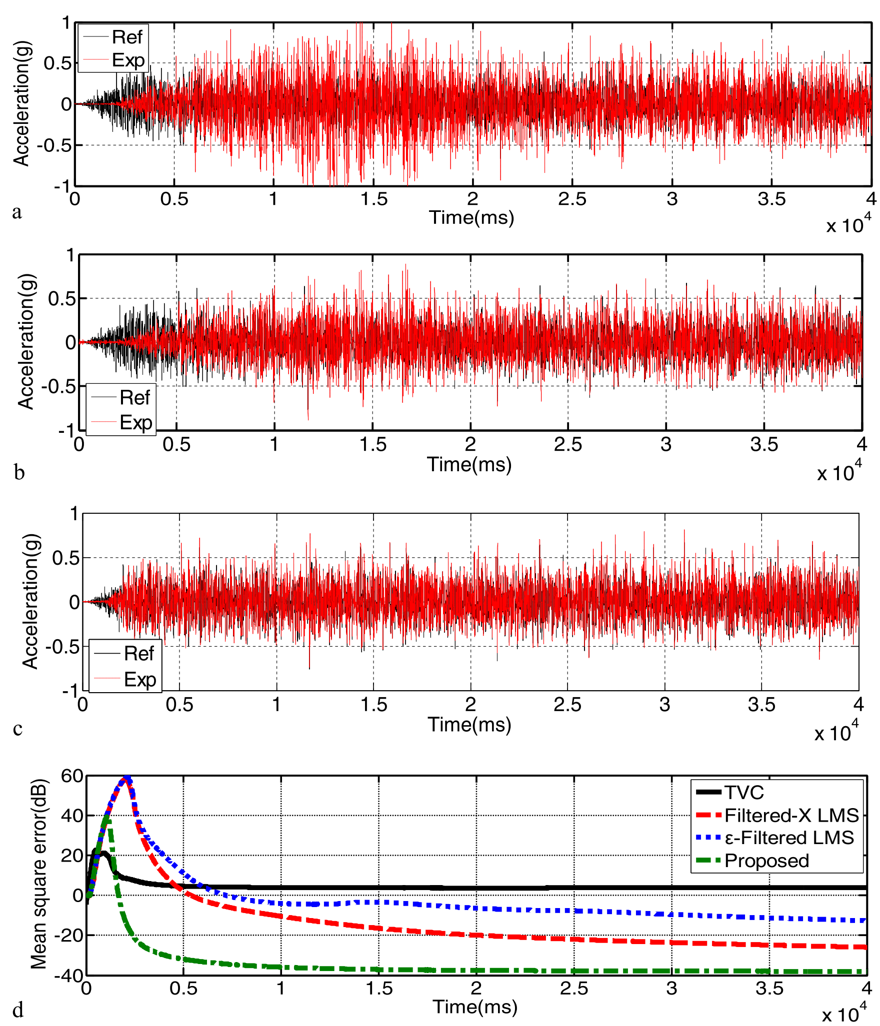

Fig. 15. Experimental results using the acceleration random signal with the frequency range of 2–50 Hz: (a) the Filtered-X LMS algorithm, (b) the ε-Filtered LMS algorithm, and (c) the proposed controller, and (d) their mean-square errors.

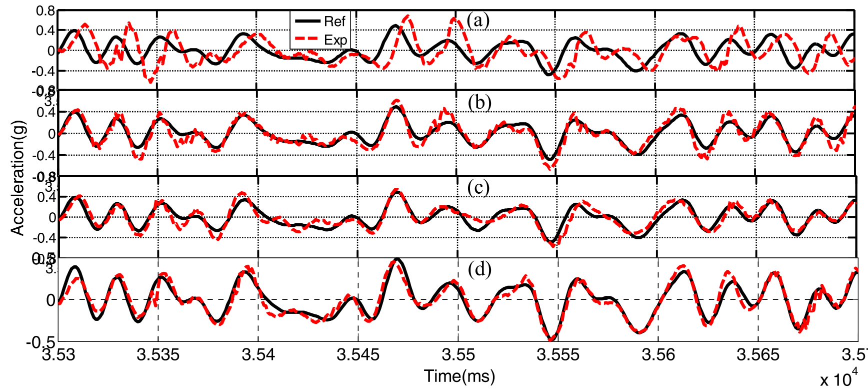

Fig. 16. Experimental results using the acceleration random command signal with the frequency range of 2–50 Hz: (a) the TVC, (b) the Filtered-X LMS algorithm, (c) the ε-Filtered LMS algorithm, and (d) the proposed controller.

Aiming at verifying the convergence performance of the proposed controller while the gain or the signal is changed or switched, the 6-DOF EHST is first excited by a random acceleration reference signal with the frequency range of 2–60 Hz and the relatively small amplitude to obtain a convergence weight, and the desired acceleration signal is then switched to an earthquake signal to investigate control performances of the proposed controller using the previous experimental convergence weight. Experimental results with different algorithms are shown in Fig. 17(a–d) and the power spectrum density of the reference earthquake signal and the output response signal are shown in Fig. 17(e), from which it can be seen that the convergence weight adapted for the 2–60 Hz random signal can also be employed for the earthquake signal when the frequency range is within 2–60 Hz without another iteration, and the acceleration tracking performance on the power spectrum density for the earthquake signal is more perfect than that with the TVC, and the Filtered-X LMS and ε-Filtered LMS algorithms, which indicates effectiveness of the proposed controller.

Finally, six random acceleration reference excitation signals with the amplitude of 1 g and the frequency range of 2–80 Hz are applied on six DOFs to demonstrate effectiveness of the proposed controller. Experiment results of six DOFs have been compared and similar improvements are found in Fig. 18. It can be seen that the proposed controller yields a magnitude and phase frequency bandwidth ($\pm$ 4 dB point and $-$ 90°) well over 70 Hz, and the worst case overturning sensitivity is 4 dB and $-$ 50° phase delay in six DOFs. The TVC has a magnitude and phase frequency bandwidth ($\pm$ 4 dB point and $-$ 90°) about 60 Hz and 15 Hz in six DOFs, respectively. These experimental results show that when the servo controller with the tuned TVC gains shown in Table 4 does not yield reasonable performances, the proposed controller can modify online dynamic characteristics of the 6-DOF EHST with better acceleration tracking performances.

From the above experimental results, it can be concluded that the proposed controller yields suitable acceleration tracking performances for different desired acceleration reference signals. Hence, it might be considered as a suitable solution for different electro-hydraulic servo system applications.

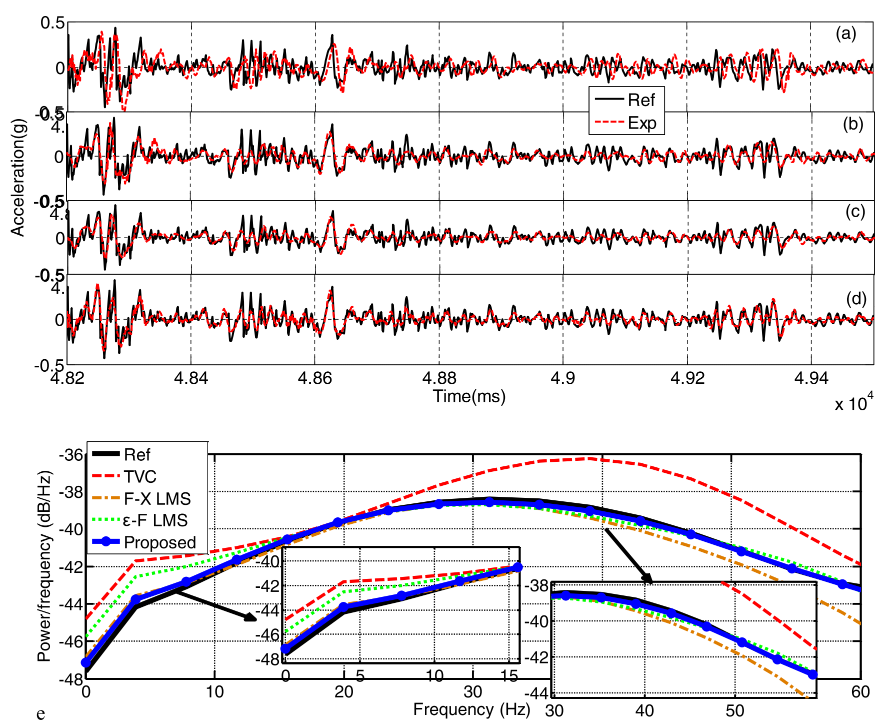

Fig. 17. Earthquake signal experimental results: (a) the TVC, (b) the Filtered-X LMS algorithm, (c) the ε-Filtered LMS algorithm, and (d) the proposed controller, and (e) their power-frequency.

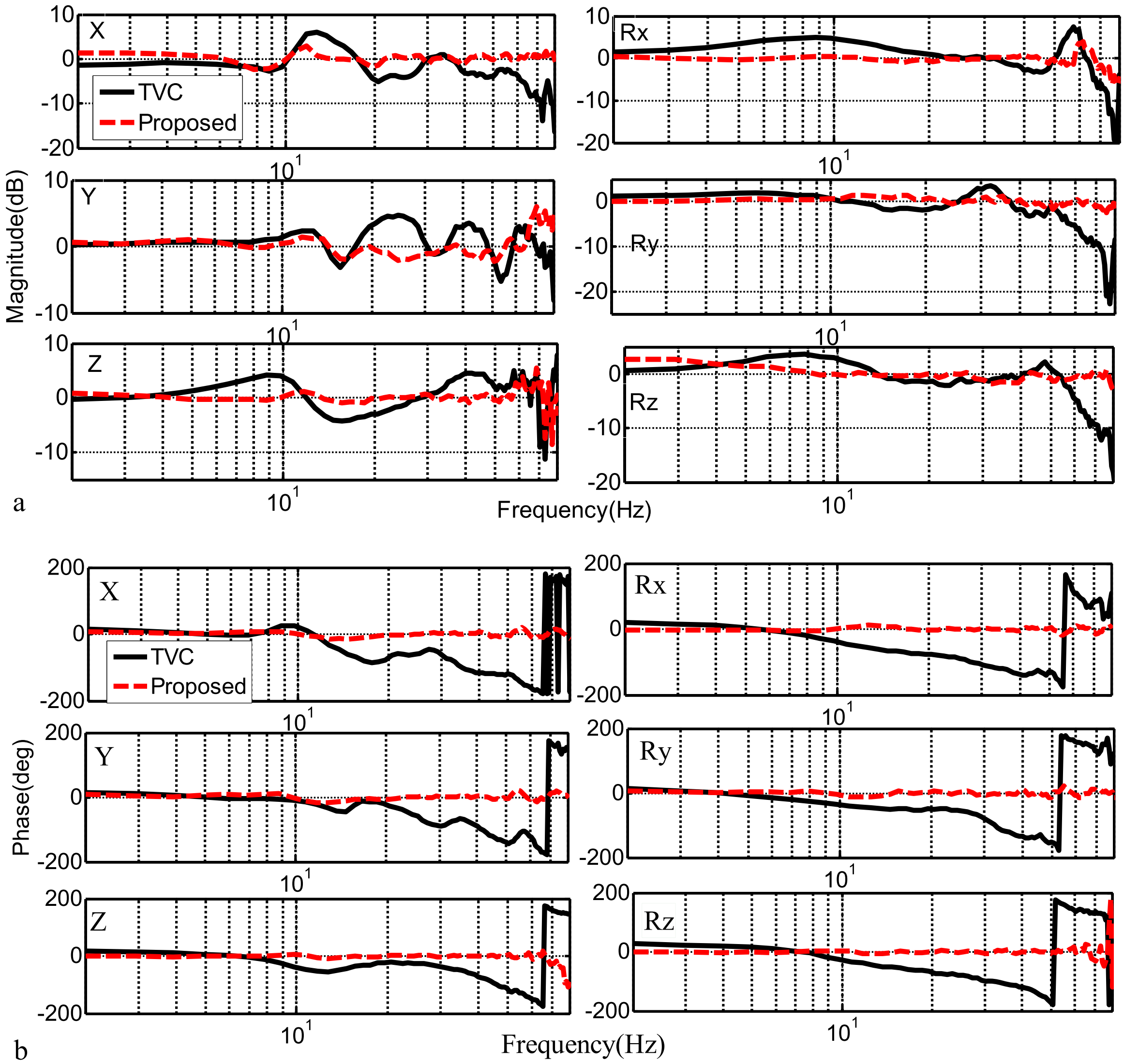

Fig. 18. Frequency characteristics of experimental results with the TVC and the proposed controller using 2–80 Hz acceleration random reference signal for the 6-DOF EHST: (a) magnitude and (b) phase.

6. Conclusions

A novel combined controller is developed and successfully applied to a 6-DOF EHST for improving acceleration tracking performances. The proposed controller incorporates:

- a PID controller,
- a coordinate controller with two transform matrices,
- a TVC including a TVC-FB and a TVC-FF,
- an OIMC based on an identified transfer function by the RELS identification algorithm,
- a zero-phase-error tracking controller for a non-minimum-phase system,
- an improved internal model controller to cancel out modeling error,
- a delay compensator for improving the delay of hydraulic actuators,
- an online adaptive controller based on a LMS algorithm.

The coordinate controller is the most important controller that includes two transform matrices and the PID controller, which makes the 6-DOF EHST normally work. The most popular controller applied in the 6-DOF EHST is the PID controller, but it can only improve acceleration tracking accuracy in a very limited range because hydraulic position servo systems regularly exhibit a poor damping ratio and a lower frequency bandwidth. The TVC is a commonly used controller applied in the 6-DOF EHST because it can extend the frequency bandwidth of the acceleration closed-loop and increase the damping ratio and stability of the 6-DOF EHST. The OIMC significantly improves acceleration tracking accuracy, but the stable inverse model must be designed because the identified acceleration closed-loop transfer function is a non-minimum-phase system and its inverse model is unstable. The adaptive controller based on the LMS algorithm yields a high-quality acceleration tracking waveform after it converges to its optimal solution, but they can exhibit poor transient response when the frequency bandwidth of the acceleration command signal exceeds the phase frequency bandwidth of the acceleration closed-loop.

The proposed controller combines advantages of these control technologies; so it is complex compared to the PID controller or the TVC. Nevertheless, the controller can be easily tuned using the following method. The coordinate controller and the PID controller are first tuned by means of two transform matrices in (3) and (4) and the position step output response, respectively. The TVC-FB is then tuned to improve stability of the EHST by means of the position step output. After the position closed-loop system is tuned by the TVC-FB, the TVC-FF is tuned to extend the frequency bandwidth of the acceleration closed-loop by means of real-time display of the magnitude and phase characteristics of the acceleration closed-loop. A complete servo controller for the 6-DOF EHST is implemented. A reasonable tracking performance of an acceleration command waveform can then be guaranteed. The OIMC and the delay compensator can extend the frequency bandwidth, especially the phase frequency bandwidth, of the acceleration closed-loop by means of off-line designed methods using experimentally measured acceleration data from the tuned servo control system. The online adaptive controller is finally employed to obtain a high-fidelity acceleration waveform if tuned dynamic characteristics of the 6-DOF EHST are changed during real-time testing under disturbances. Thus, the proposed controller can accelerate the rate of convergence of the LMS algorithm by means of the designed off-line compensators and yield high-fidelity acceleration tracking accuracy after the LMS algorithm converges to its optimal solution.

A 6-DOF EHST based on MATLAB/xPC rapid prototyping is employed to verify effectiveness of the proposed controller and experimental results show that the proposed controller yields a better acceleration tracking performance with respect to different acceleration signals and six DOFs. The proposed control is effective not only for different electro-hydraulic servo systems, but also for other servo control systems.

Acknowledgments

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Appendix A. Open-loop dynamic model of the EHST

A single hydraulic actuator configuration of the EHST is established in Fig. 3, where Q_1 is flow into the chamber and Q_2 is flow out of the chamber of the hydraulic cylinder, respectively, which can be defined by [38]

$$Q_1 = C_d w x_v \sqrt{2(p_s - p_1)/\rho}, \tag{A1}$$

$$Q_2 = C_d w x_v \sqrt{2p_2/\rho}, \tag{A2}$$

where x_v is the spool displacement of the servo-valve, p_1 and p_2 are pressures in the two chambers, C_d is the discharge coefficient, w is the constant area gradient of servo-valve orifices, ρ is the mass density of the fluid, p_s is the hydraulic supply pressure, and p_L is the load pressure defined by $p_L = p_1 - p_2$. By neglecting the impact of leak on the hydraulic actuator and compression of oil, the linearized load flow Q_L can be obtained by

$$Q_L = K_q x_v - K_c P_L, \tag{A3}$$

where K_q and K_c are the linearized flow gain and the flow pressure coefficient, respectively, which can be defined by $K_q = \partial Q_L/x_v = C_d w\sqrt{(p_s - p_L)/\rho}$ and $K_c = \partial Q_L/\partial p_L = C_d w x_v\sqrt{(p_s - p_L)/\rho}/2(p_s - p_L)$. The load flow Q_L from the servo-valve into two actuator chambers is expressed by [39]

$$Q_L = A_p\frac{dx_p}{dt} + C_{tp}p_L + \frac{V_t}{4\beta_e}\cdot\frac{dp_L}{dt}, \tag{A4}$$

where A_p is the effective area of the hydraulic actuator, x_p is the piston displacement, dx_p/dt is the velocity of the actuator piston, V_t is the total volume of the actuator, β_e is the effective bulk modulus, and C_{tp} is the total leakage coefficient of the actuator including an internal leakage coefficient C_{ip} and an external leakage coefficient C_{ep}, whose relation is given by $C_{tp} = C_{ip} + C_{ep}/2$. By neglecting the mass of oil and the nonlinear load such as dry friction and the dynamics of the piston, the load pressure p_L is described by Newton's second law [40]

$$p_L = \frac{1}{A_p}\left(m_t\frac{d^2x_p}{dt^2} + B_c\frac{dx_p}{dt}\right), \tag{A5}$$

where m_t is the total mass of the actuator piston, the load, and the specimen, B_c is the viscous damping coefficient between the piston and the payload. The relationship between the control signal i and the position x_v of the servo-valve spool is expressed by [3]

$$G_{sv}(s) = \frac{X_v(s)}{I(s)} = \frac{K_{sv}}{\left(\frac{s^2}{\omega_{sv}^2} + \frac{2\xi_{sv}}{\omega_{sv}}s + 1\right)}, \tag{A6}$$

where K_{sv}, ω_{sv} and ξ_{sv} are the gain, the angular natural frequency and the damping coefficient of the servo-valve, respectively.

Appendix B. Transfer function identification

In order to design the OIMC, the estimated actual acceleration closed-loop system transfer function needs to be identified and its closed-loop model can be regarded as an auto-regressive moving average [41]; its differential equation is given by

$$y(k) = -\sum_{i=1}^{n_a} A_i y(k-i) + \sum_{i=1}^{n_b} B_i u(k-i) + \sum_{i=1}^{n_d} D_i v(k-i) + v(k), \tag{B1}$$

where k is the time index, $y(k)$ and $u(k)$ are the system output and input vectors, respectively, $v(k)$ is the white noise vector with zero mean, $A(z)$, $B(z)$, and $D(z)$ are matrix-coefficient polynomials, which can be expressed by $A(z) = 1 + A_1z^{-1} + A_2z^{-2} + \cdots\cdots + A_{n_a}z^{-n_a}$, $B(z) = B_1z^{-1} + B_2z^{-2} + \cdots\cdots + B_{n_b}z^{-n_b}(n_a \ge n_b)$, $D(z) = 1 + D_1z^{-1} + D_2z^{-2} + \cdots\cdots + D_{n_d}z^{-n_d}(n_a \ge n_d)$, with z^{-1} is a unit delay operator $z^{-1}y(k) = y(k-1)$. An identification formulated model can be obtained from (B1):

$$y(k) = \theta^T\varphi(k) + v(k), \tag{B2}$$

where $\theta^T = [A_1, A_2, \cdots\cdots A_{n_a}, B_1, B_2, \cdots\cdots B_{n_b}, D_1, D_2, \cdots\cdots D_{n_d}]$, and

$$\begin{aligned}\varphi(k) = [&-y^T(k-1), \cdots\cdots, -y^T(k-n_a), -u^T(k-1), \cdots\cdots,\\ &-u^T(k-n_b), -v^T(k-1), \cdots\cdots, -v^T(k-n_d)].\end{aligned}$$

Employing the estimated value $\hat{v}(k)$ to replace $v(k)$ because $v(k)$ in $y(k)$ cannot be measured with sensors, one has

$$\hat{v}(k) = y(k) - \hat{y}(k) = y(k) - \hat{\varphi}^T(k)\hat{\theta}, \tag{B3}$$

where $\hat{\theta}^T = [\hat{A}_1, \hat{A}_2, \cdots\cdots \hat{A}_{n_a}, \hat{B}_1, \hat{B}_2, \cdots\cdots \hat{B}_{n_b}, \hat{D}_1, \hat{D}_2, \cdots\cdots \hat{D}_{n_d}]$, and $\hat{\varphi}(k) = [-y^T(k-1), \cdots\cdots, -y^T(k-n_a), -u^T(k-1), \cdots\cdots, -u^T(k-n_b), -\hat{v}^T(k-1), \cdots\cdots, -\hat{v}^T(k-n_d)]$

One can employ $\hat{\varphi}(k)$ to replace $\varphi(k)$ and the RELS algorithm to identify the acceleration closed-loop model, which is given by [25]

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \hat{\varphi}^T(k)\hat{\theta}(k-1)], \tag{B4}$$

$$K(k) = \frac{P(k-1)\hat{\varphi}(k)}{\rho + \hat{\varphi}^T(k)P(k-1)\hat{\varphi}(k)}, \tag{B5}$$

$$P(k) = \frac{1}{\rho}[I - K(k)\hat{\varphi}^T(k)]p(k-1), \tag{B6}$$

where ρ is a forgetting factor whose value range is $0 < \rho < 1$, $K(k)$ is the gain vector at step k, and $P(k)$ is the inverse correlation matrix at step k and I is an identity matrix.

Appendix C. Convergence analysis

By substituting (22) into (28), the LMS algorithm is written as

$$\begin{aligned} w^*(k+1) &= w^*(k) + \rho z^{-L}x(k)x^T(k) - \rho\frac{(1+\Delta G)w^*(k)}{1+\alpha\Delta G}x(k)x^T(k) - \rho\frac{1-\alpha}{1+\alpha\Delta G}n(k)x^T(k) \\ &= [I - \rho\frac{(1+\Delta G)}{1+\alpha\Delta G}x(k)x^T(k)]w^*(k) + \rho z^{-L}x(k)x^T(k) - \rho\frac{1-\alpha}{1+\alpha\Delta G}n(k)x^T(k). \end{aligned} \tag{C1}$$

Subtracting $\frac{(1+\Delta G)}{1+\alpha\Delta G}z^{-L}$ from both sides of (C1), one has

$$V(k+1) = w^*(k+1) - \frac{(1+\Delta G)}{1+\alpha\Delta G}z^{-L}, \tag{C2}$$

$$V(k) = w^*(k) - \frac{(1+\Delta G)}{1+\alpha\Delta G}z^{-L}. \tag{C3}$$

Taking expectation at both sides of (C3) yields

$$E[V(k+1)] = \{I - E[\rho]\frac{(1+\Delta G)}{1+\alpha\Delta G}R\}E[V(k)]. \tag{C4}$$

The following two inequalities can be obtained:

$$\frac{\mu}{\frac{\beta}{\lambda + \sum_{i=k-l+1}^{k} e^2(k)} + (1-\beta)\|x(k)\|^2} \ge \frac{\mu}{\frac{\beta}{\lambda} + (1-\beta)\|x(k)\|^2}, \tag{C5}$$

$$E[\rho] \ge \frac{\mu}{\frac{\beta}{\lambda} + (1-\beta)E[\|x(k)\|^2]} = \frac{\mu}{\frac{\beta}{\lambda} + (1-\beta)R}. \tag{C6}$$

Substituting (C6) into (C4) yields

$$E[V(k+1)] \le \{I - \frac{\mu}{(\beta/\lambda + (1-\beta)R)}\frac{(1+\Delta G)}{(1+\alpha\Delta G)}R\}E[V(k)]. \tag{C7}$$

Decomposing $\mathbf{R}$ as $\mathbf{R} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$, where $\boldsymbol{\Lambda}$ is a diagonal matrix of eigenvalues of $\mathbf{R}$ containing positive real-valued eigenvalues λ_i of the reference [42]. $E[V'_{k,i}]$ can be expressed in terms of the initial condition $V'_{0,i}$ [38]

$$E[V'_{k,i}] \le [1 - \frac{\mu}{(\beta/\lambda + (1-\beta)R)}\frac{(1+\Delta G)}{(1+\alpha\Delta G)}\lambda_i]^k V'_{0,i}. \tag{C8}$$

The above equation shows that the i_{th} component of the mean weight vector $E[V'_{k,i}]$ is a geometric sequence with geometric ratio $1 - \mu(\beta/\lambda + (1-\beta)R)(1+\Delta G)/(1+\alpha\Delta G)\lambda_i$. The mean weight vector $E[w(k)]$ converges to Wiener solution when the following equation is satisfied:

$$\left|1 - \frac{\mu}{(\beta/\lambda + (1-\beta)R)}\frac{(1+\Delta G)}{(1+\alpha\Delta G)}\lambda_i\right| < 1, (i = 1, \cdots N). \tag{C9}$$

References

[1] Kenta S, Makoto I, Motohiro K, et al. Adaptive compensation for reaction force with frequency variation in shaking table systems. IEEE Trans Ind Electron 2009;56(10):3864–71.
[2] Lee SK, Park EC, Min KW. Real-time hybrid shaking table testing method for the performance evaluation of a tuned liquid damper controlling seismic response of building structures. J Sound Vib 2007;302:596–612.
[3] Shen G, Zhu ZC, Li X, et al. Real-time electro-hydraulic hybrid system for structural testing subjected to vibration and force loading. Mechatronics 2016;33:49–70.
[4] Shen G, Zhu ZC, Zhang L, et al. Adaptive feed-forward compensation for hybrid control with acceleration time waveform replication on electro-hydraulic shaking table. Control Eng Pract 2013;8(21):1128–42.
[5] Plummer AR. Model-based motion control for multi-axis servohydraulic shaking tables. Control Eng Pract 2016;53:109–22.
[6] Plummer AR. A general co-ordinate transformation framework for multi-axis motion control with application in the testing industry. Control Eng Pract 2010;10:598–607.
[7] Shen G, Zhu ZC, Tang Y, et al. Combined control strategy using internal model control and adaptive inverse control for electro-hydraulic shaking table. IMechE C: J Mech Eng Sci 2013;227(10):2348–60.
[8] Yao JJ, Di DT, Jiang GL, et al. Acceleration amplitude-phase regulation for electro-hydraulic servo shaking table based on LMS adaptive filtering algorithm. Int J Control 2012;85(10):1581–92.
[9] Xu Y, Hua HX, Han JW. Modeling and controller design of a shaking table in an active structural control system. Mech Syst Signal Process 2008;2:1917–23.
[10] Tagawa Y, Kajiwara K. Controller development for the E-Defense shaking table. IMechE I: J Syst Control Eng 2007;221(2):171–81.
[11] Plummer AR. Feedback linearization for acceleration control of electro-hydraulic actuators. IMechE I: J Syst Control Eng 1997;211:139–69.
[12] Yao JY, Jiao ZX, Ma DW. Extended-state-observer-based output feedback nonlinear robust control of hydraulic systems with backstepping. IEEE Trans Ind Electron 2014;61:6285–93.
[13] Yao JY, Jiao ZX, Ma DW, Yan L. High-accuracy tracking control of hydraulic rotary actuators with modeling uncertainties. IEEE/ASME Trans Mechatron 2014;19(2):633–41.
[14] Wang CW, Jiao ZX, Wu S, et al. Nonlinear adaptive torque control of electro-hydraulic load system with external active motion disturbance. Mechatronics 2014;24(1):32–40.
[15] Yao JJ, Wang XC, Hu SH, et al. Adaline neural network-based adaptive inverse control for an electro-hydraulic servo system. J Vib Control 2011;17(13):2007–14.
[16] Shen G, Zheng ST, Ye ZM, et al. Adaptive inverse control of time waveform replication for electro hydraulic shaking table. J Vib Control 2011;17:1611–33.
[17] Stoten DP, Shimizu N. The feedforward minimal control synthesis algorithm and its application to the control of shaking-tables. IMechE I: J Syst Control Eng 2007;221(3):423–44.
[18] Stoten DP, Eduardo GG. Adaptive control of shaking tables using the minimal controller synthesis algorithm. Philos Trans R Soc A 2001;359:1697–723.
[19] Shen G, Lv GM, Ye ZM, et al. Implementation of electrohydraulic shaking table controllers with a combined adaptive inverse control and minimal control synthesis algorithm. IET Control Theory Appl 2011;5(13):1471–83.
[20] Shen G, Zhu ZC, Zhao JS, et al. Real-time tracking control of electro-hydraulic force servo systems using offline feedback control and adaptive control. ISA Trans 2017;67:356–70.
[21] Gizatullin AO, Edge KA. Adaptive control for a multi-axis hydraulic test rig. IMechE I: J Syst Control Eng 2006;221(2):183–98.
[22] Karshenas AM, Dunnigan MW, Williams BW. Adaptive inverse control algorithm for shock testing. IEE Control Theory Appl 2000;147(3):267–76.
[23] Uchiyama Y, Mukai M, Fujita M. Robust control of electro dynamic shaker with 2dof control using H∞ filter. J Sound Vib 2009;326:75–87.
[24] Vasilis KD, Harris PM, Ioannis NP. On the acceleration-based adaptive inverse control of shaking tables. Earthq Eng Struct Dyn 2015;44:1329–50.
[25] Vaudrey MA, Baumann WT, Saunders WR. Stability and operating constraints of adaptive LMS-based feedback control. Automatica 2003;39(4):595–605.
[26] Chen C, Ricles JM, Sause R, et al. Experimental evaluation of an adaptive inverse compensation technique for real-time simulation of a large-scale magneto-rheological fluid damper. Smart Mater Struct 2010;19:1–12.
[27] Karer G, Mušič G, Igor Š. Feedforward control of a class of hybrid systems using an inverse model. Math Comput Simulat 2011;82:414–27.
[28] Brian PR, Lucy YP, Dale AL. Nonminimum phase dynamic inversion for settle time applications. IEEE Trans Control Syst Technol 2009;17(5):989–1005.
[29] Mercan O. Analytical and experimental studies on large scale, real-time pseudo dynamic testing (Ph.D. dissertation).Bethlehem: Dep of Civil and Environmental Engineering, Lehigh University; 2007.
[30] Chen C, Ricles JM. Stability analysis of explicit integration algorithms with actuator delay for real-time hybrid testing. Earthq Eng Struct Dyn 2008;37(4):597–613.

[31] Horiuchi T, Inoue M, Konno T, et al. Real-time hybrid experimental system with actuator delay compensation and its application to a piping system with energy absorber. Earthq Eng Struct Dyn 1999;28(10):1121–41.
[32] Horiuchi T, Konnno T. A new method for compensation actuator delay in real-time hybrid experiment. Philos Trans R Soc Lond A 2001;59(1786):1893–909.
[33] Mahnaz H. Adaptive control for a class of MIMO nonlinear time delay systems against time varying actuator failures. ISA Trans 2015;57:23–42.
[34] Wang YJ. Characterization and quenching of friction-induced limit cycles of electro-hydraulic servovalve control systems with transport delay. ISA Trans 2010;49:489–500.
[35] Joris DC, Michel V, Jan S. Off-line feed-forward and H∞ feedback control on a vibration rig. Control Eng Pract 2003;11:129–40.
[36] Gregory LP. Adaptive inverse control of linear and nonlinear systems using dynamic neural networks. IEEE Trans Neural Netw 2003;14(2):360–76.
[37] Widrow B, Walach E. Adaptive inverse control. Wiley-IEEE Press; 2008.
[38] Lin TL, Huang WP, Ren HL, et al. New compound energy regeneration system and control strategy for hybrid hydraulic excavators. Autom Constr 2016;68:11–20.
[39] Liu Y, Gong GF, Yang HY, et al. Regulating characteristics of new tamping device exciter controlled by rotary valve. IEEE/ASME Trans Mechatron 2016;21(1):497–505.
[40] Shen W, Mai YF, Su XY, et al. A new electric hydraulic actuator adopted the variable displacement pump. Asian J Control 2016;18(1):178–91.
[41] Muhammad SA. Maximum likelihood least squares identification method for active noise control systems with autoregressive moving average noise. Automatica 2016;69:1–11.
[42] August K, Daniel VG. On the use of a priori knowledge in adaptive inverse control. IEEE Trans Circuit Syst-I: Fundam Theory Appl 2000;47(1):54–62.